

Stability of solutions to Heat/Diffusion equations

Consider two solutions to the IVP for Heat/Diff eq.

$$\begin{cases} \partial_t u_1 - k \partial_{xx} u_1 = 0 & x \in [0, L] \quad t > 0 \\ u_1(x, 0) = \phi_1(x) \\ u_1(0, t) = g_1(t) \quad u_1(L, t) = h_1(t) \end{cases}$$

$$\begin{cases} \partial_t u_2 - k \partial_{xx} u_2 = 0 \\ u_2(x, 0) = \phi_2(x) \\ u_2(0, t) = g_2(t), \quad u_2(L, t) = h_2(t) \end{cases}$$

(2)

We assume closeness of ϕ_1 to ϕ_2 , g_1 to g_2 and h_1 to h_2
and WTS that u_1 is close to u_2
want to show

Consider $w = u_1 - u_2$ which now solves

$$\left\{ \begin{array}{l} \partial_t w - k \partial_{xx} w = 0 \quad x \in [0, L] \quad t \geq 0 \\ w(x, 0) = \phi_1(x) - \phi_2(x) \quad x \in [0, L] \\ w(0, t) = g_1(t) - g_2(t); \quad w(L, t) = h_1(t) - h_2(t) \end{array} \right.$$

By MAX/MIN Principle imply:

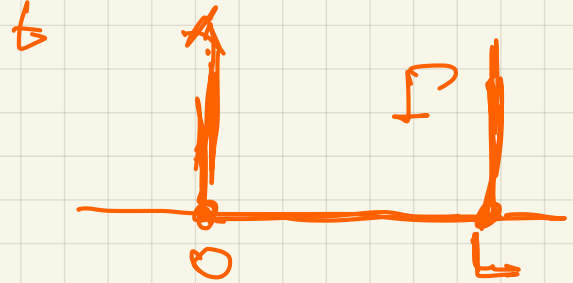
$$\min_{(x,t) \in \Gamma} \{ w(x,t) \} \leq \underbrace{w(x,t)}_{(x,t) \in R} \leq \max_{(x,t) \in \Gamma} \{ w(x,t) \}$$

□

In fact we have (from MAX/MIN principle).

(3)

$$\max_{(x,t) \in \bar{I}} \{ |w(x,t)| \} \leq \max_{(x,t) \in R} \{ w(x,t) \} \leq \max_{(x,t) \in I} \{ |w(x,t)| \}$$



Then:

$$\max_{(x,t) \in R} \{ |u_1(x,t) - u_2(x,t)| \} \leq \max_{(x,t) \in I} \{ |w(x,t)| \}$$

$$= \max_{\substack{0 \leq x \leq L \\ t > 0}} \{ |\phi_1(x) - \phi_2(x)|, |g_1(t) - g_2(t)|, |h_1(t) - h_2(t)| \}$$

$$t \in [0, T]$$

④

Hence if the maximum ^(in x) of $|\phi_1(x) - \phi_2(x)|$;
and the maximum (in t) of $|g_1(t) - g_2(t)|$ and of
 $|h_1(t) - h_2(t)|$ are small then maximum of
 $(x, t) \in \mathbb{R}$
 $|u_1(x, t) - u_2(x, t)|$ is also small.

Smallness here is understood to hold UNIFORMLY
in the (x, t) variables.

An alternative topology to measure stability
which is given by square integrability and proved

via the energy functional - $E[w](t)$ for Heat/Diffusion ^⑤

Claim (to be proved) is decreasing in t :

$$0 \leq E[w](t) \leq E[w](0) \text{ for any}$$

a solution to Heat/Diffusion equation.

Note then that w also satisfies the same equation

$$0 \leq \boxed{E[w](t) \leq E[w](0)}^* \text{ for } t \geq 0.$$

But this implies that the energy of the solution

is controlled by $\underbrace{E[\phi_1 - \phi_2]}_{\text{number}}$

In fact we'll prove that the energy for heat/bff. ⁽⁶⁾

eq. is
$$E[w](t) = \frac{1}{2} \int_0^L |w(x,t)|^2 dx. \quad (+)$$

and $\frac{dE}{dt} \leq 0$ so energy decreases

Then (*) says

$$\int_0^L |u_1(x,t) - u_2(x,t)|^2 dx \leq \underbrace{\int_0^L |\phi_1(x) - \phi_2(x)|^2 dx}_{\text{number}}$$

This type of stability is
called in the square integral sense

if $|\phi_1 - \phi_2|$ is
square integrable

Energy Functional for Heat/Diffusion

7

Where does $(+)$ come from and why it is decreasing in t .

w solves

$$0 = w_t - k w_{xx}$$

$$w \cdot 0 = (w_t - k w_{xx}) w$$

$$= 0$$

$(+)$

$$\begin{cases} w_t - k w_{xx} = 0 \\ w(x, 0) = 0 \\ w(0, t) = 0 \\ w(L, t) = 0 \end{cases}$$

$x \in [0, L]$

$$0 = w w_t - k w w_{xx}$$

Integrate both sides.

$$0 = \int_0^L w w_t dx - k \int_0^L w w_{xx} dx \stackrel{\text{IBP}}{=} \int_0^L w w_t dx - k \int_0^L w w_{xx} dx$$

$$= \frac{1}{2} \int_0^L \frac{\partial}{\partial t} |w(x,t)|^2 dx - k \left[w w_x \right]_{x=0}^{x=L} - \int_0^L (w_x)^2 dx \quad (8)$$

Hence

$$\frac{1}{2} \frac{\partial}{\partial t} \int_0^L |w(x,t)|^2 dx = \underbrace{- \int_0^L (w_x)^2 dx}_{\leq 0}$$

So if we call

$$E[w](t) = \frac{1}{2} \int_0^L |w(x,t)|^2 dx \quad \text{and}$$

(9)

w satisfies $(++)$ then $\frac{dE}{dt} \leq 0$ so

Energy $E[w](t)$ decreases. $E[w](t) \leq E[w](0)$

This proves our earlier claim

Uniqueness of solutions to the IVP for heat

$$\begin{aligned} \text{(IVP)} \quad & \left\{ \begin{aligned} u_t - k u_{xx} &= 0 & x \in [0, L] \\ u(x, 0) &= \phi(x) & x \in [0, L] \\ u(0, t) &= g(t) \quad u(L, t) = h(t) \end{aligned} \right. \end{aligned}$$

Then if u_1 and u_2 solve IVP for heat

Then $w = u_1 - u_2$ solves $(++) \Rightarrow$

Since $E[w](t) \leq E[w](0)$

$$= \frac{1}{2} \int \underbrace{|w(x,0)|^2}_{=0} dx$$

As $\downarrow$

$$0 \leq \frac{1}{2} \int_0^L \underbrace{|w(x,t)|^2}_{\geq 0} dt \leq 0 \text{ for all } t \geq 0.$$

$$\Rightarrow |w(x,t)| = 0 \text{ for all } 0 \leq x \leq L \text{ and all } t \geq 0.$$

$$\therefore w(x,t) = u_1(x,t) - u_2(x,t) = 0 \text{ for all } x \in [0, L], t \geq 0.$$

$$\text{Hence } u_1(x,t) = u_2(x,t) \text{ for all } x \in [0, L], t \geq 0$$

So we get uniqueness as desired *